

Lab Mid Preparation Questions

Part 1: Arithmetic & Logic Challenges

1. 3-Bit Subtractor with Magnitude Check

Design a circuit that takes two 3-bit numbers, A and B, and performs $A - B$.

Only show result if $A \geq B$; otherwise output 000.

2. Even/Odd Summation Indicator

Add two 2-bit numbers. Output 3-bit sum and a 4th bit = 1 if sum is even.

3. Conditional 2's Complementer

For 3-bit input A: if $M=1$ output 2's complement, else output A.

4. 4-Bit Range Detector

Output 1 only if input is between 5 (0101) and 10 (1010).

5. 2-Bit Multiplier via Addition

Multiply two 2-bit numbers using AND gates and a 7483 adder.

Part 2: MUX, Encoder, Decoder Challenges

6. MUX-Based Full Adder

Implement Full Adder using 74153 MUX.

7. 4-Bit Priority Bit Counter

Use 74148 to find highest 1 bit.

8. Divisibility by 4 Checker

Use decoder; output 1 if number ends in 00.

9. Modified BCD to Excess-3

If input ≤ 4 add 3, else subtract 1.

10. Universal Gate Secret Key

Use only NOR gates to check equality of two 2-bit inputs.

Preparation Checklist:

- Memorize IC pinouts
- Understand active-low logic
- Know subtractor logic using XOR and carry-in